

P2.12.26 证明: 循环群一定是阿贝尔群

设  $G = \langle a \rangle$  是循环群.

则对  $\forall x_1 = a^i \in G, x_2 = a^j \in G$

$$x_1 x_2 = a^{i+j} = a^{j+i} = x_2 x_1$$

因此  $G$  是阿贝尔群.

反之不成立 例如 Klein 四元群  $\{a, b, c, e\}$

它是阿贝尔群但不是循环群

P2.27.15  $L$  为格.  $\forall a_1, a_2, \dots, a_n \in L$

如果  $a_1 \wedge a_2 \wedge \dots \wedge a_n = a_1 \vee a_2 \vee \dots \vee a_n$

证明  $a_1 = a_2 = \dots = a_n$

$$\underbrace{a_1 \wedge a_2 \wedge \dots \wedge a_n}_{\text{LHS}} \leq a_1 \leq \underbrace{a_1 \vee a_2 \vee \dots \vee a_n}_{\text{RHS}}$$

又  $\text{LHS} = \text{RHS}$

$$\text{则 } a_1 = \text{LHS} = \text{RHS}$$

$$\text{同理可得 } a_i = \text{LHS} = \text{RHS} \quad i=2, \dots, n$$

$$\text{故 } a_1 = a_2 = \dots = a_n$$

井论第

P22T6  $L$  是格  $a, b \in GL$  且  $a \leq b \leq c$

$$\text{证明 } a \vee b = b \wedge c$$

$$\text{由 } a \leq b \text{ 得 } a \vee b = b$$

$$b \leq c \text{ 得 } b \wedge c = b$$

$$\text{故 } a \vee b = b \wedge c \quad \# \text{ 证毕}$$

补充题: 设  $G$  为群.  $a, b \in G$ . 且  $ab = ba$

$$\text{若 } |a| = n, |b| = m, \quad \langle a \rangle \cap \langle b \rangle = \{e\}$$

$$\text{则 } |ab| = [m, n]$$

$$\text{证: 设 } |ab| = r \quad m \mid r \quad n \mid r \quad (m, n) = 1$$

$$\text{则 } r = [m, n]$$

利用  $r \mid [m, n]$  和  $[m, n] \mid r$

$$\textcircled{1} r \mid [m, n]$$

$$(ab)^{[m, n]} = a^{[m, n]} b^{[m, n]} = a^{k_1 n} b^{k_2 m} = e \cdot e = e$$

$$\text{故 } r \mid [m, n]$$

②  $[m, n] \mid r$

$$a^r = (ab)^r (b^{-1})^r = e \cdot (b^{-1})^r = (b^{-1})^r \in \langle a \rangle$$

$$\text{又 } \because (b^{-1})^r \in \langle b \rangle \text{ 且 } \langle a \rangle \cap \langle b \rangle = \{e\}$$

$$\text{故 } (b^{-1})^r = e \text{ 即 } b^r = e$$

$$\text{又 } \because |a| = m, |b| = n$$

$$\text{故 } m \mid r \text{ 且 } n \mid r$$

$$\text{因此 } [m, n] \mid r$$

综合①②知  $[m, n] = r$  # 证毕